Reaction	Product	Name of the reaction
$CH_3CH_2OH + SOCl_2 \xrightarrow{\text{pyridine}} ?$	--------	--------
$CH_3CH_2Br + AgF \longrightarrow ?$	--------	--------
$C_6H_5Cl + Na \xrightarrow{\text{ether}} ?$	--------	--------

40. Discuss the aromatic nucleophilic substitutions reaction of chlorobenzene.

41. Account for the following

    (i) t-butyl chloride reacts with aqueous KOH by S_N1 mechanism while n-butyl chloride reacts with S_N2 mechanism.

    (ii) p-dichloro benzene has higher melting point than those of o-and m-dichloro benzene.

42. In an experiment ethyliodide in ether is allowed to stand over magnesium pieces. Magnesium dissolves and product is formed

    a) Name the product and write the equation for the reaction.

    b) Why all the reagents used in the reaction should be dry? Explain

    c) How is acetone prepared from the product obtained in the experiment.

43. Write a chemical reaction useful to prepare the following:

    i) Freon-12 from Carbon tetrachloride

    ii) Carbon tetrachloride from carbon disulphide

44. What are Freons? Discuss their uses and environmental effects

45. Predict the products when bromoethane is treated with the following

    i) KNO_2

    ii) $AgNO_2$

46. Explain the mechanism of S_N1 reaction by highlighting the stereochemistry behind it

47. Write short notes on the the following

    i) Raschig process

    ii) Dows Process

    iii) Darzens process

48. Starting from CH_3MgI, How will you prepare the following?

    i) Acetic acid

    ii) Acetone

    iii) Ethyl acetate

    iv) Iso propyl alcohol

    v) Methyl cyanide

49. Complete the following reactions

    i) $CH_3 - CH = CH_2 + HBr \xrightarrow{\text{Peroxide}}$

    ii) $CH_3 - CH_2 - Br + NaSH \xrightarrow[H_2O]{\text{alcohol}}$

    iii) $C_6H_5Cl + Mg \xrightarrow{THF}$

    iv) $CHCl_3 + HNO_3 \xrightarrow{\Delta}$

    v) $CCl_4 + H_2O \xrightarrow{\Delta}$

50. Explain the preparation of the following compounds

    i) DDT ii) Chloroform

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